

Exam Date & Time: 02-Jun-2023 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FOURTH SEMESTER B.TECH END SEMESTER EXAMINATIONS, MAY-JUNE 2023

PYTHON PROGRAMMING [CSE 4309]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

Instructions to Candidates: Answer ALL questions Missing data may be suitably assumed

- 1) Justify Python syntax to create a tuple with a single element in comparison to the syntax of expressions. Use an example to clarify the differences. Why this problem cannot exist for a tuple with more than one element? (3)
 - A)
 - B) Give two features of a set data type in Python which are distinct from a list data type. Write a Python program to remove duplicate elements of a list l1 using a set and store the result into a new list l2. Use built-in functions and show output. (4)
 - C) Why a list of characters is not considered same as a string. Justify by taking an example. Given an input sentence, write a Python program to break the sentence into words and to convert a word into its characters using built-in functions and write output. (3)
- 2) Calculate rainfall in gallons for some number of inches on 1 acre. Write a python program to do the following.
 - A)
 - i. Prompt the user for the number of inches that have rain fallen.
 - ii. Find the volume (in cubic feet) of water (where volume = depth * area).
 - iii. Convert the volume (in cubic feet) to gallons. (3)Consider the conversion formulas:
1 acre = 43,560 square feet
1 cubic foot = 7.48051945 gallons.
 - B) Write a non-function Python program to check whether the given number is Armstrong number or not using while loop? (4)
[Hint: 371 is an Armstrong number since $3^3 + 7^3 + 1^3 = 371$].
 - C) Write a Python Program to do the following List operations. (3)

	0	1	2
0	2	5	8
1	3	7	4
2	1	6	9
3	4	2	0

- i) Create the following using a simple 2D list named as 'L1' using the standard Python indexing:
- ii) Using the 2D list 'L1', ask the user which row they would like displayed and display just that row. Ask them to enter a new value and add it to the end of the row using built-in function and display the row again.
- 3) Give examples for built-in functions and type conversion functions. Demonstrate how multiple values can be returned from a function as a list of values to the main program. (3)
- A)
- B) Illustrate the four different ways in which importing can be done? Give syntax and usage of each one. (3)
- C) With a neat diagram, show the workflow required to add a widget for creating a basic GUI window. Using this workflow, write a complete Python program to design a button widget followed by a label widget both packed to the left using a pack system call. Display the design. (4)
- 4) Write a python program that has a class Student with Name, Roll no, and marks of three subjects. Create a method called display that prints the student information and total marks in three subjects (3)
- A)
- B) Why numpy array is preferred over list? Explain any three numpy functions with example. (3)
- C) Using Sqlite Module create a table named Books with columns Title, Author, and Year of Publication. Insert three books information into Books table. Fetch year and Title from the Books table in year wise sorted order. (4)
- 5) Write a Python program to combine each line from first file with the corresponding line in second file. (3)
- A)
- B) What is a dictionary? Write a python program to sort (ascending order) a dictionary by value. (3)
- C) What is cloud computing? Illustrate the levels of cloud computing services? (4)

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